

Motor Duty Class

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Motor Duty Class

- Now a days, in almost every applications, [electric motors](#) are used, and to control them [electrical drives](#) are employed.
- But the operating time for all motors are not the same.
- Some of the motors runs all the time, and some of the motor's run time is shorter than the rest period.

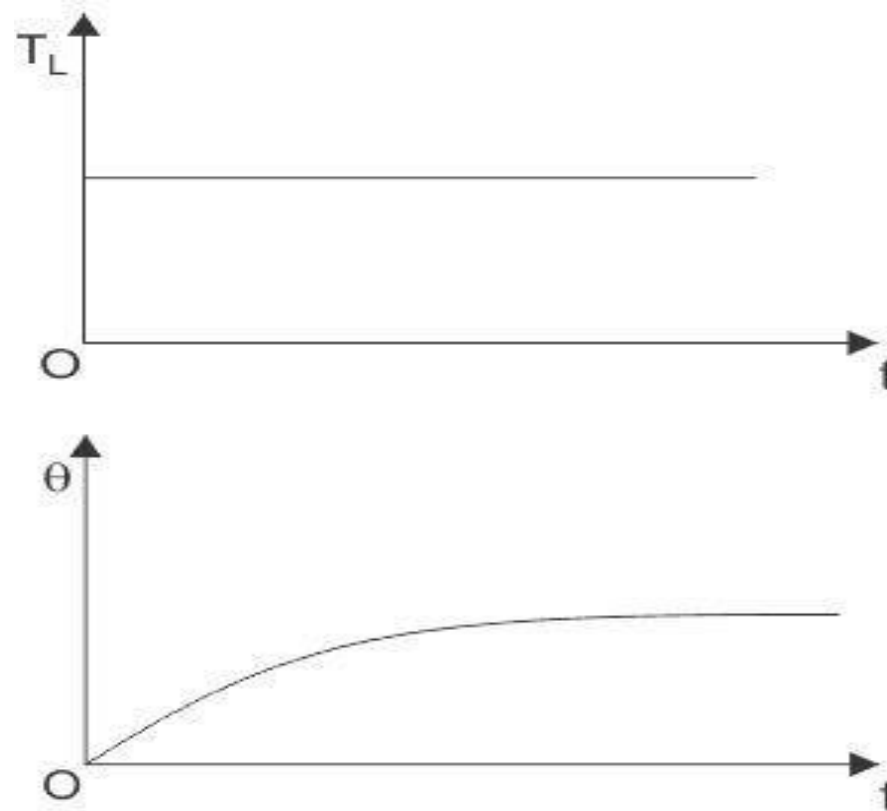
Motor Duty Class

► Concept of **motor duty class** is introduced and on the basis of this duty cycles of the motor can be divided in eight categories such as,

1. Continuous duty
2. Short time duty
3. Intermittent periodic duty
4. Intermittent periodic duty with starting
5. Intermittent periodic duty with starting and braking
6. Continuous duty with intermittent periodic loading
7. Continuous duty with starting and braking
8. Continuous duty with periodic speed changes

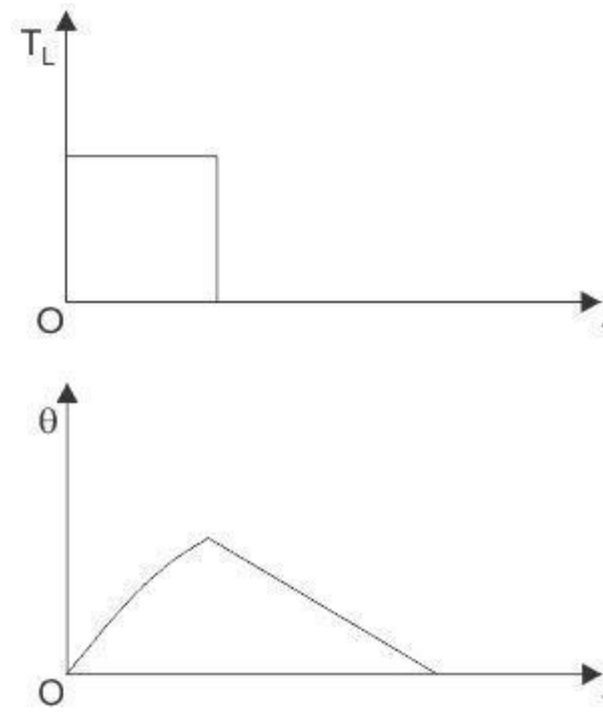
Continuous Duty Cycle

- ▶ This duty denotes that, the motor is running long enough and the electric motor temperature reaches the steady state value. These motors are used in paper mill drives, compressors, conveyors etc.



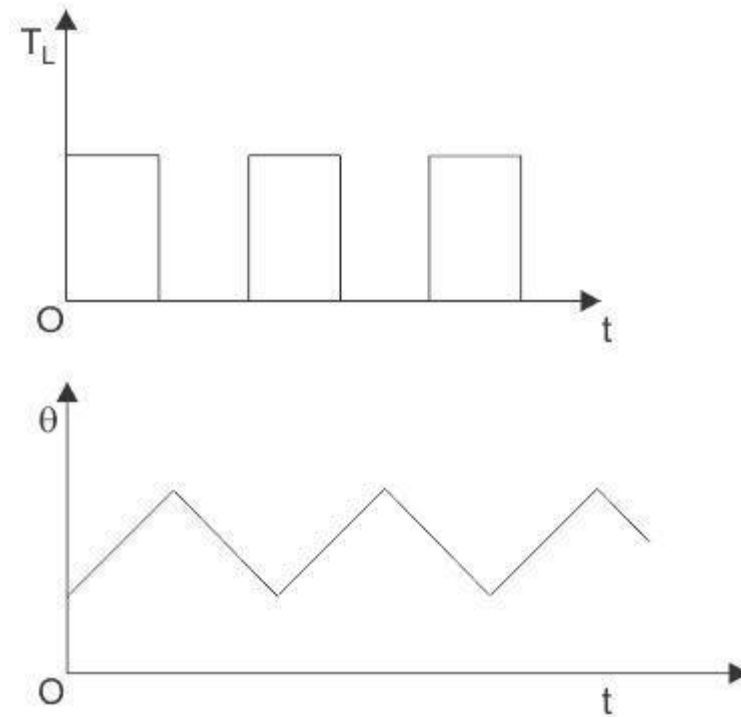
Short Time Duty Cycle

- ▶ In these motors, the time of operation is very low and the heating time is much lower than the cooling time. So, the motor cools off to ambient temperature before operating again. These motors are used in crane drives, drives for house hold appliances, valve drives etc.



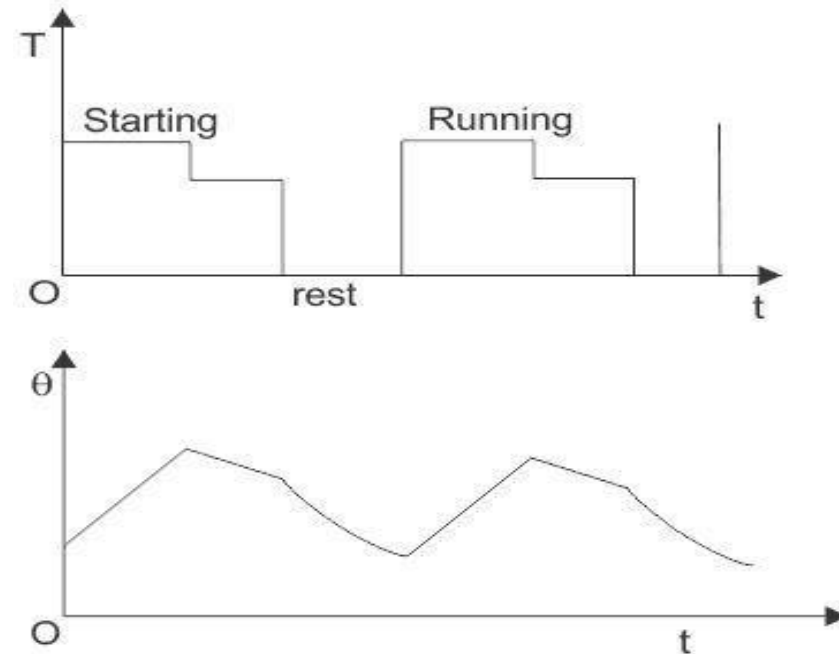
Intermittent Periodic Duty

- Here the motor operates for some time and then there is rest period. In both cases, the time is insufficient to raise the temperature to steady state value or cool it off to ambient temperature. This is seen at press and drilling machine drives.



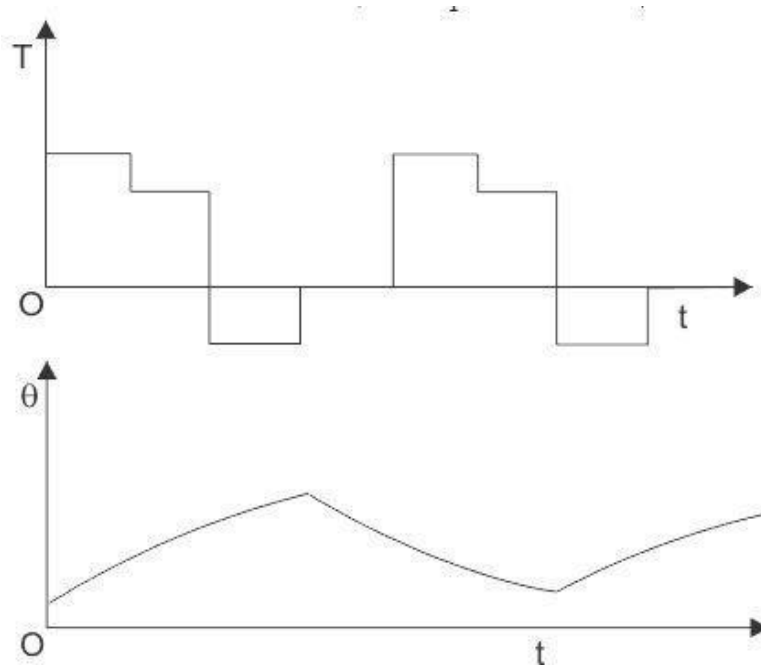
Intermittent Periodic Duty With Starting

- ▶ In this type of duty, there is a period of starting, which cannot be ignored and there is a heat loss at that time. After that there is running period and rest period which are not adequate to attain the steady state temperatures. This motor duty class is widely used in metal cutting and drilling tool drives, mine hoist etc.



Intermittent Periodic Duty With Starting & Breaking

- In this type of drives, heat loss during starting and braking cannot be ignored. So, the corresponding periods are starting period, operating period, braking period and resting period, but all the periods are too short to attain the respective steady state temperatures, these techniques are used in billet mill drive, manipulator drive, mine hoist etc



Classes of Motor Duty

- Continuous Duty with Intermittent Periodic Loading: In this type of motor duty, everything is same as the periodic duty but here a no load running period occurs instead of the rest period. Pressing, cutting are the examples of this system.
- Continuous Duty with Starting and Braking: It is also a period of starting, running and braking and there is no resting period. The main drive of a blooming mill is an example.
- Continuous Duty with Periodic Speed Changes: In this type of motor duty, there are different running periods at different loads and speeds. But there is no rest period and all the periods are too short to attain the steady state temperatures.

Motor De-rating

- Any adverse operating conditions require that the motor performance be de-rated. These conditions include; ambient temperature above 40°C, motor mounting position, drive switching frequency or the drive being oversized for the motor
- Ambient Temperatures: The ambient temperature around the motor must be taken into account. For ambient temperatures above 40°C the torque must be de-rated.
- Mounting Arrangement: The motor torque must be de-rated if the motor mounting surface is heated from an external source, such as a gearbox. The motor is connected to a poor thermal conductor. The motor is mounted with the connectors on the side or vertical. The motor is in a confined space with restricted air flow.

Energy Efficient Motor

- Energy efficient motors use less electricity, run cooler and often last longer than NEMA (National Electrical Manufacturers Association) B motors of the same size.
- To effectively evaluate the benefits of high efficiency electric motors, we must define "efficiency". For an electric motor, efficiency is the ratio of mechanical power delivered by the motor (output) to the electrical power supplied to the motor (input).
- $\text{Efficiency} = (\text{Mechanical Power Output} / \text{Electrical Power Input}) \times 100\%$

Energy Efficient Motor

- Thus, a motor that is 85% efficient converts 85% of the electrical energy input into mechanical energy
- The remaining 15% of the electrical energy is dissipated as heat, evidenced by a rise in motor temperature
- Energy efficient electric motors utilize improved motor design and high quality materials to reduce motor losses, therefore improving motor efficiency
- The improved design results in less heat dissipation and reduced noise output

Motor Nameplate



Voltage

3~MOT MG 90SA2-24FF165-C2			
50 Hz	P ₂ 1,50 kW	No 85807906	
	U 220-240D/380-415Y		V
Eff. % 82	I _{1/1} 5.90/3.40	A	
	I _{max} 6.50/3.75	A	
n 2860-2890 min ⁻¹		cos φ 0.85-0.79	
CL F	IP 55		0346
DE 6305.2Z.C4 NDE 6205.2Z.C3			
			
			
GRUNDFOS 			
Made in Hungary			

85807906

Frequency



Phase

3~ MOT MG 90SA2-24FF165-C2			
50 Hz	P ₂ 1,50 kW		No85807906
	U 220-240D/380-415Y		V
Eff. % 82	I _{1/1} 5.90/3.40		A
	I _{max} 6.50/3.75		A
n 2860-2890 min ¹ cos φ 0.85-0.79			
CL F	IP 55		0346
DE 6305.2Z.C4 NDE 6205.2Z.C3			
			
			
GRUNDFOS 			
Made in Hungary			

Power Factor

3~MOT MG 90SA2-24FF165-C2			
50 Hz	P ₂ 1,50 kW	No85807906	
	U 220-240D/380-415Y	V	
Eff. % 82	I _{1/1} 5.90/3.40	A	
	I _{max} 6.50/3.75	A	
n 2860-2890 min ⁻¹	cosφ 0.85-0.79		
CL F	IP 55		0346
DE 6305.2Z.C4 NDE 6205.2Z.C3			
			
  			
Made in Hungary			

85807906

Full Load Speed

3~MOT MG 90SA2-24FF165-C2				85807906
50 Hz	P ₂ 1,50 kW	No85807906		
	U 220-240D/380-415Y	V		
Eff. % 82	I _{1/1} 5.90/3.40		A	
	I _{max} 6.50/3.75		A	
n 2860-2890 min ⁻¹		cos φ 0.85-0.79		
CL F	IP 55	0346		
DE 6305.2Z.C4 NDE 6205.2Z.C3				
				
  				
Made in Hungary				

Duty

CE		GRUNDFOS®				EFF 2	
TYPE MMG132S-2-38FF265-E		CAT.NO. 340333060		PART NO. 83315217			
5.5 kW		Lw _{50Hz} 80	dB(A)	MAX.AMB. 40 °C		INS. F	CONN. Δ
ENCL. IP55		EFF.(100%FL) 85.7%		EFF.(75%FL) 86%			
HZ 50		HZ 60		DUTY S1		TP111	
VOLT. 380-415Δ/660-690Y		VOLT. 380-480Δ/660-690Y		WGT. 66		kg	
AMP. 11 / 6.4		AMP. 10.5-8.6/6.1-5.0		Δ Y W2 U2 V2 U1 V1 W1			
R.P.M. 2900-2920		R.P.M. 3470-3525					
COSφ 0.89-0.86		COSφ 0.92-0.88					
BRG.D.E. 6208ZZ		N.D.E. 6208ZZ					
YEAR 2001 WEEK 28		SER.NO. 0001					
MADE IN CHINA				6314			

References

- ▶ <https://www.electrical4u.com/motor-duty-class/>